

JavaScript Array Methods Complete Guide

Definition + Syntax + Pseudo code + Working + Output

push()

Definition: Adds element at end

Syntax: `arr.push(item)`

Pseudo Code: Take item -> add end

Code: `let a=[1,2];a.push(3);console.log(a)`

Output: `[1,2,3]`

pop()

Definition: Removes last element

Syntax: `arr.pop()`

Pseudo Code: Remove last item

Code: `let a=[1,2,3];a.pop();console.log(a)`

Output: `[1,2]`

shift()

Definition: Removes first element

Syntax: `arr.shift()`

Pseudo Code: Remove first item

Code: `let a=[1,2,3];a.shift()`

Output: `[2,3]`

unshift()

Definition: Adds first element

Syntax: `arr.unshift(item)`

Pseudo Code: Add at start

Code: `let a=[2,3];a.unshift(1)`

Output: `[1,2,3]`

length

Definition: Returns total items

Syntax: `arr.length`

Pseudo Code: Count items

Code: `console.log([1,2,3].length)`

Output: 3

slice()

Definition: Returns selected items

Syntax: `arr.slice(start,end)`

Pseudo Code: Take indexes

Code: `console.log([1,2,3,4].slice(1,3))`

Output: `[2,3]`

splice()

Definition: Add/remove items

Syntax: `arr.splice(index,delete,item)`

Pseudo Code: Modify original array

Code: `let a=[1,2,3];a.splice(1,1)`

Output: `[1,3]`

concat()

Definition: Join arrays

Syntax: `arr.concat(arr2)`

Pseudo Code: Merge arrays

Code: `console.log([1,2].concat([3]))`

Output: `[1,2,3]`

includes()

Definition: Checks value

Syntax: `arr.includes(2)`

Pseudo Code: Search value

Code: `console.log([1,2,3].includes(2))`

Output: `true`

indexOf()

Definition: Find index

Syntax: `arr.indexOf(item)`

Pseudo Code: Find first index

Code: `console.log([10,20].indexOf(20))`

Output: `1`

join()

Definition: Convert array to string

Syntax: `arr.join('-')`

Pseudo Code: Join elements

Code: `console.log([1,2,3].join('-'))`

Output: 1-2-3

reverse()

Definition: Reverse array

Syntax: `arr.reverse()`

Pseudo Code: Reverse elements

Code: `console.log([1,2,3].reverse())`

Output: [3,2,1]

sort()

Definition: Sort array

Syntax: `arr.sort()`

Pseudo Code: Arrange items

Code: `console.log([3,1,2].sort())`

Output: [1,2,3]

map()

Definition: Creates new array

Syntax: `arr.map(cb)`

Pseudo Code: Loop and modify

Code: `console.log([1,2].map(x=>x*2))`

Output: [2,4]

filter()

Definition: Returns matching values

Syntax: `arr.filter(cb)`

Pseudo Code: Loop->condition->save true

Code: `console.log([10,20,30].filter(x=>x>15))`

Output: [20,30]

find()

Definition: Returns first matching value

Syntax: `arr.find(cb)`

Pseudo Code: Find first condition

Code: `console.log([10,20].find(x=>x>10))`

Output: 20

findIndex()

Definition: Returns first index

Syntax: `arr.findIndex(cb)`

Pseudo Code: Find index

Code: `console.log([10,20].findIndex(x=>x>10))`

Output: 1

forEach()

Definition: Loops array

Syntax: `arr.forEach(cb)`

Pseudo Code: Loop each item

Code: `[1,2].forEach(x=>console.log(x))`

Output: 1 2

some()

Definition: Checks any condition true

Syntax: `arr.some(cb)`

Pseudo Code: Check one true

Code: `console.log([1,2].some(x=>x>1))`

Output: true

every()

Definition: Checks all true

Syntax: `arr.every(cb)`

Pseudo Code: Check all true

Code: `console.log([2,4].every(x=>x>1))`

Output: true

reduce()

Definition: Reduce to one value

Syntax: `arr.reduce(cb)`

Pseudo Code: Combine values

Code: `console.log([1,2,3].reduce((a,b)=>a+b))`

Output: 6

flat()

Definition: Flatten nested array

Syntax: `arr.flat()`

Pseudo Code: Remove nesting

Code: `console.log([1,[2]].flat())`

Output: `[1,2]`